MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

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VECTOR ANALYSIS

(Vector in two-dimensions)

Lec # 1

Vector Analysis

Vector in two-dimensions

Basic Definitions

1) SCALAR

A quantity which is completely specified by its magnitude is called a " SCALAR "

Number value with units

Scalar Example	Magnitude
Speed	35 m/s
Distance	25 meters
Age	16 years

2) VECTOR

A physical quantity which has both magnitude and direction is called a " VECTOR "

Vector Example	Magnitude and Direction
Velocity	35 m/s, North
Acceleration	10 m/s^2, South
Displacement	20 m, East

3) REPRESENTATION OF A VECTOR

A vector is represented by <u>directed line segment</u>

For Example: $\overline{AB}$ represents a vector whose magnitude

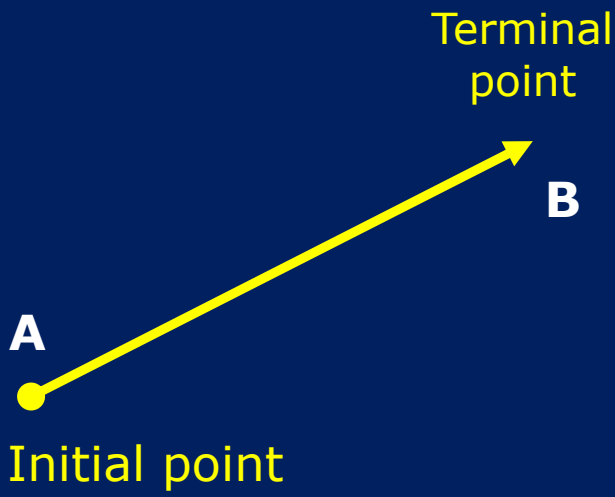
denote by $|\overline{AB}|$ is the length AB and its direction is from **A** to **B**



An arrow over a letter $\vec{A}$

Or

a letter in bold face **A**



4) COMPONENTS OF VECTORS

If $\vec{v} = a\hat{\imath} + b\hat{\jmath}$

the vectors $a\hat{\imath}$ and $b\hat{\jmath}$ are the vectors components of $\vec{v}$ in the direction of $\hat{\imath}$ and $\hat{\jmath}$.

The number a and b are the scalar components of $\vec{v}$ in the direction of $\hat{\imath}$ and $\hat{\jmath}$.

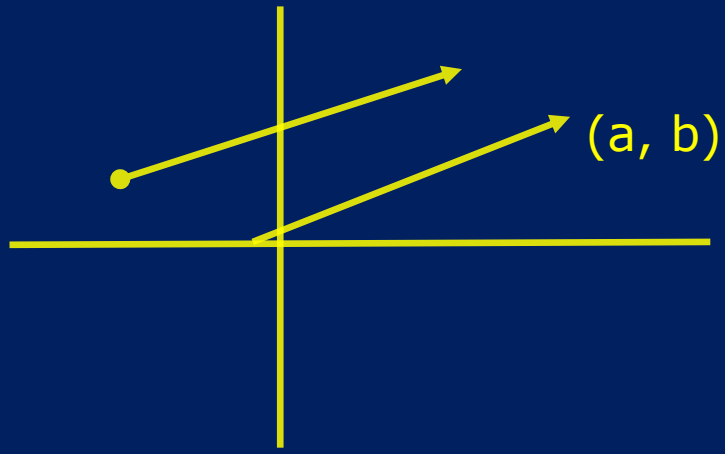
5) EQUAL VECTORS / EQUALITY OF VECTORS

Two Vectors are equal if they have same magnitude and have the same or parallel directions.

Algebraic Definition

If $\vec{v} = a\hat{\imath} + b\hat{\jmath}$ and

$\vec{v}_1 = \boldsymbol{a_1}\hat{\imath} + \boldsymbol{b_1}\hat{\jmath}$ then



$$\vec{v} = \vec{v}_1$$

$$a\hat{\imath} + b\hat{\jmath} = \boldsymbol{a_1}\hat{\imath} + \boldsymbol{b_1}\hat{\jmath}$$

$$\Leftrightarrow$$

$$a = \boldsymbol{a_1} \text{ and } b = \boldsymbol{b_1}$$

Two vectors are equal $\Leftrightarrow$ their scalar components in the direction of $\hat{i}$ and $\hat{j}$ are identical.

REMARK

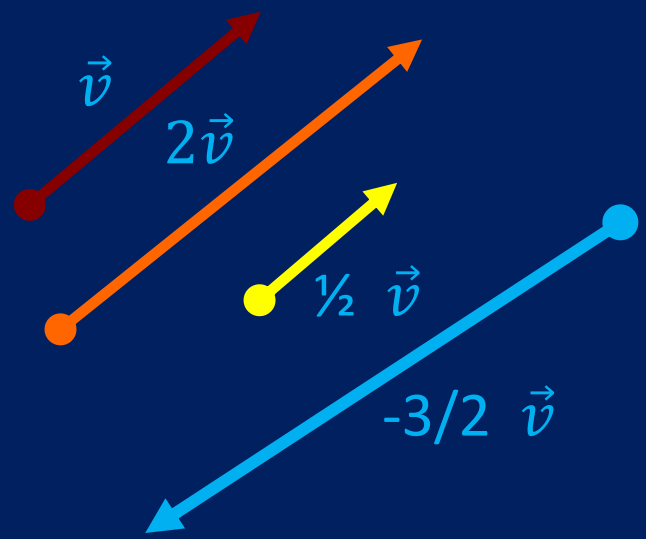
A Vector may be translated from one location to another, provided neither the magnitude nor the direction is changed.

6) SCALAR MULTIPLICATION OF $\vec{v}$

If c is a Scalar and $\vec{v}$ is a vector, then $c\vec{v}$ is defined as.

"A vector whose magnitude is |C| times the magnitude of $\vec{v}$ and whose direction is either the same as that of $\vec{v}$ (if C > 0) or opposite that of $\vec{v}$ (if C < 0) "

Illustrations are given in Figures. We refer to $c\vec{v}$ as a scalar multiple of $\vec{v}$.



SCALAR MULTIPLICATION

Scalar multiplication can be accomplished component by components

If C is a scalar and $\vec{v} = a\hat{\imath} + b\hat{\jmath}$ is a vector, then

$$c\vec{v} = c\left(a\hat{\imath} + b\hat{\jmath}\right)$$

$$= (ca)\hat{\imath} + (cb)\,\hat{\jmath}$$

REMARK

The length of $c\vec{v}$ is |c| times the length of $\vec{v}$

$$|\, c\vec{v}\,| = |(ca)\hat{\imath} + (cb)\,\hat{\jmath}\,| = \sqrt{c^2a^2 + c^2b^2}$$

$$= \sqrt{c^2(a^2 + b^2)} = \sqrt{c^2}\,\sqrt{(a^2+b^2)} = |\, c\,|\,|\,\vec{v}\,|$$

For Example

If $c = -2$ and $\vec{v} = -3\hat{\imath} + 4\hat{\jmath}$ **then**

$$|\vec{v}| = \sqrt{(-3)^2+(4)^2} = \sqrt{9+16} = \sqrt{25} = 5$$

$$\boldsymbol{c\vec{v} = (-2)(-3\hat{\imath} + 4\hat{\jmath}) = (6)\hat{\imath} - (8)\hat{\jmath}}$$

$$|\mathbf{c}\vec{v}| = \sqrt{(6)^2+(-8)^2} = \sqrt{36+64} = \sqrt{100} = 10$$

$$|\mathbf{c}| = |-2| = 2$$

$$|\boldsymbol{c\vec{v}}| = |\boldsymbol{c}|\ |\vec{v}|$$

$$10 = 2(5) = 10$$

REMARK

Non-zero vectors $\vec{v}_1$ **and** $\vec{v}_2$ have

1. the same direction if $\vec{v}_2 = c\vec{v}_1$ for some scalar $c > 0$.

2. the opposite direction if $\vec{v}_2 = c\vec{v}_1$ for some scalar $c < 0$.

7) ADDITION OF VECTORS

Algebraic Addition

Vectors may be added algebraically by adding their corresponding Scalar components.

If $\vec{v}_1 = a_1\hat{i} + b_1\hat{j}$

and $\vec{v}_2 = a_2\hat{i} + b_2\hat{j}$ then

$$\vec{v}_1 + \vec{v}_2 = (a_1 + a_2)\,\hat{i} + (b_1 + b_2)\hat{j}$$

For example: Let $\vec{v}_1 = 2\hat{i} - 4\hat{j}$ and $\vec{v}_2 = 5\hat{i} + 3\hat{j}$ then

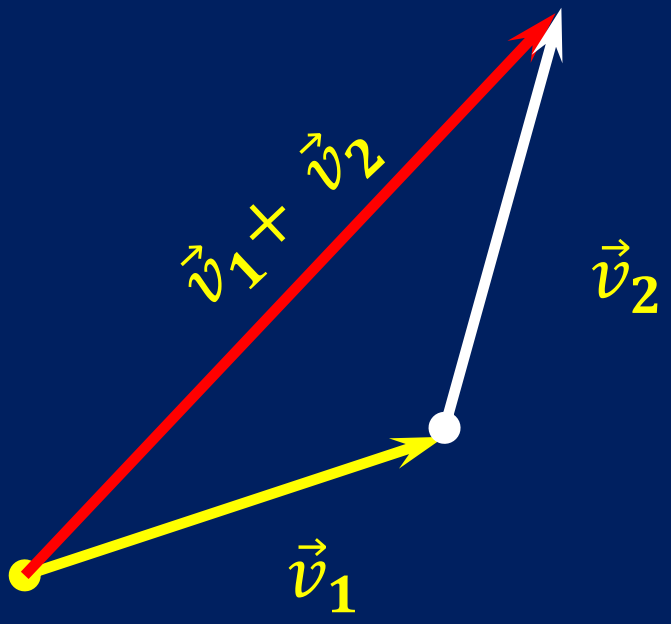
$$\vec{v}_1 + \vec{v}_2 = (2 + 5)\,\hat{i} + (-4 + 3)\hat{j}$$

$$= 7\,\hat{i} - \hat{j}$$

Graphical Addition

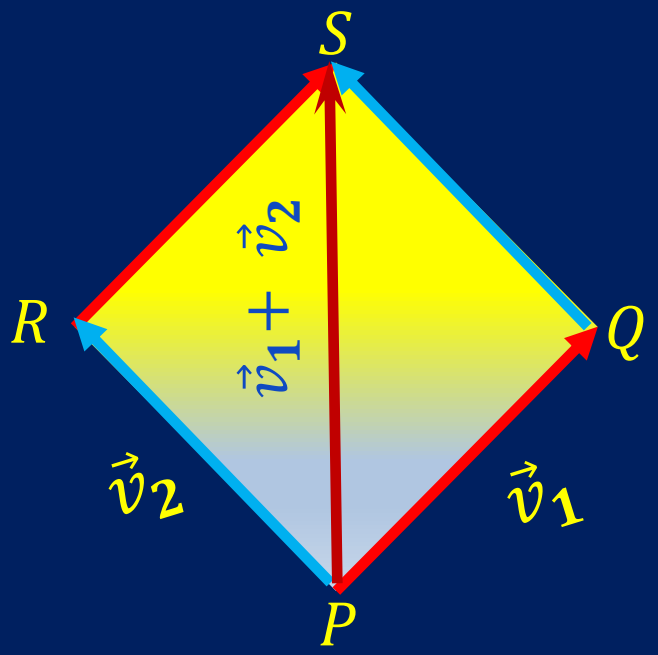
(1st method)

Two non-zero vectors $\vec{v}_1$ and $\vec{v}_2$ can be added geometrically by placing the initial point of the 2nd vector on the terminal point of the 1st vector and then drawing the line segment from the initial point of the 1st vector to the terminal point of the 2nd vector.



(2nd method)

If we construct parallelogram RPQS, then Since $\overline{PR} = \overline{QS}$, it follows that $\overline{PS} = \overline{PQ} + \overline{QS}$. If $\overline{PQ}$ and $\overline{PR}$ are two forces acting at P, then $\overline{PS}$ is the resultant forces that is, the single force that produces the same effect as the two combined forces.



8) SUBTRACTION OF VECTORS

Algebraic Subtraction

Vectors subtraction follows the algebraic law which says that corresponding scalar components are subtracted.

If $\vec{v}_1 = a_1\hat{\imath} + b_1\hat{\jmath}$

and $\vec{v}_2 = a_2\hat{\imath} + b_2\hat{\jmath}$ then

$$\vec{v}_1 - \vec{v}_2 = (a_1 - a_2)\hat{\imath} + (b_1 - b_2)\hat{\jmath}$$

Geometric Subtraction

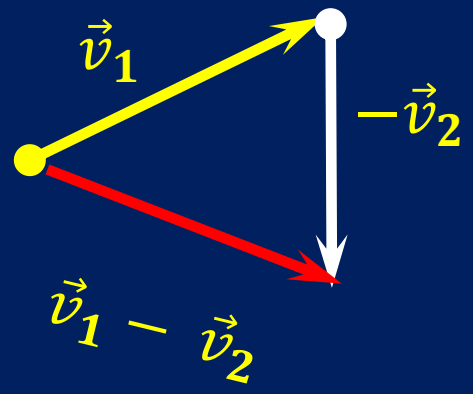
The negative of a vectors $\vec{v}$ is the vector $-\vec{v} = (-1)\vec{v}$. It has the same length as $\vec{v}$ but points in the opposite directions.

Geometric Subtraction

To subtract a vector $\vec{v}_2$ from a vector $\vec{v}_1$ we add $-\vec{v}_2$ to $\vec{v}_1$. This can be done geometrically.

1st way: **Draw as** $\vec{v}_1+(-\vec{v}_2)$

Now drawing $-\vec{v}_2$ from the tip of $\vec{v}_1$
and then drawing the vector from the initial point of $\vec{v}_1$
to the tip of $-\vec{v}_2$ as shown in Fig.

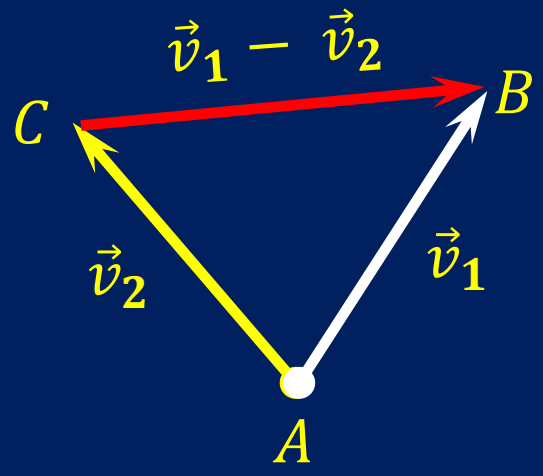


2nd way:

Draw as the vector from the tip of $\vec{v}_2$ to the tip of $\vec{v}_1$

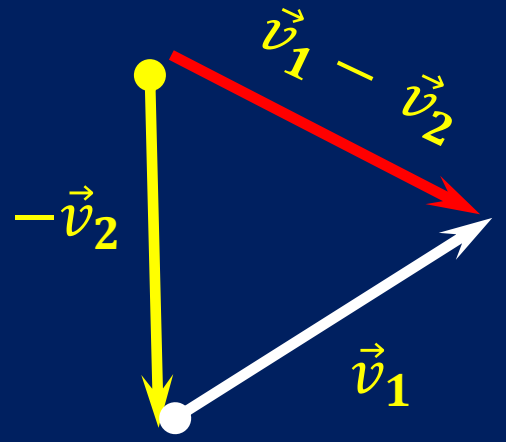
To Draw $\vec{v}_1 - \vec{v}_2$ is to draw $\vec{v}_1$ and $\vec{v}_2$ with a common initial point
and then draw $\vec{v}_1 - \vec{v}_2$ vector from the tip of $\vec{v}_2$ to the tip of $\vec{v}_1$. This is illustrated in Fig.

$$\overrightarrow{CB} = \overrightarrow{CA} + \overrightarrow{AB}$$
$$= -\vec{v}_2 + \vec{v}_1$$
$$= \vec{v}_1 - \vec{v}_2$$



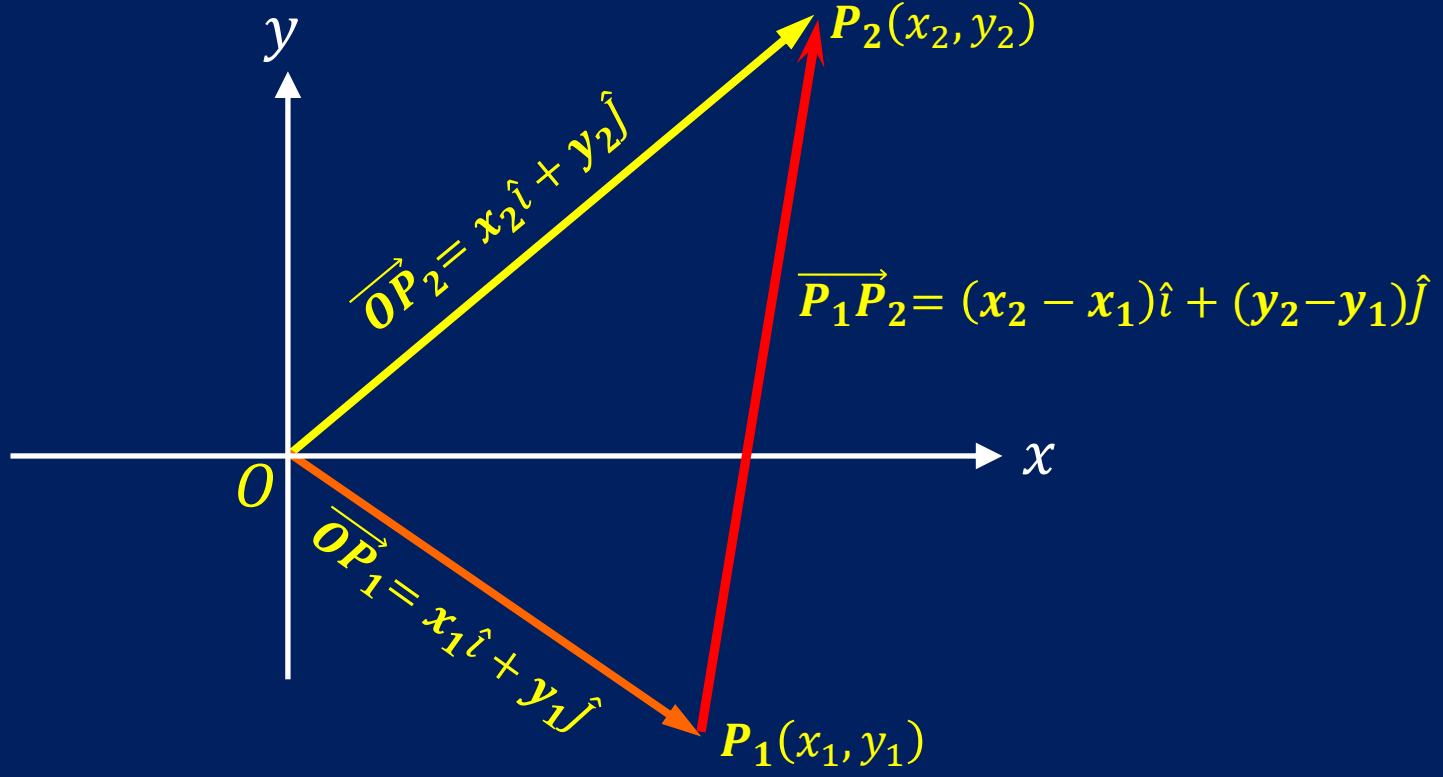
3rd way: **Draw as** $-\vec{v}_2 + \vec{v}_1$

To Draw $-\vec{v}_2 + \vec{v}_1$ is to draw $\vec{v}_1$ from the tip of $-\vec{v}_2$



9) POSITION VECTOR

To find the components of the vector from a point $P_1(x_1, y_1)$ to a point $P_2(x_2, y_2)$ by subtracting the components of $\overrightarrow{OP_1} = x_1\hat{\imath} + y_1\hat{\jmath}$ from the components of $\overrightarrow{OP_2} = x_2\hat{\imath} + y_2\hat{\jmath}$ (as shown in fig)



The vector from $\boldsymbol{P_1(x_1, y_1)}$ **to** $\boldsymbol{P_2(x_2, y_2)}$ is

$$\overrightarrow{\boldsymbol{P_1P_2}} = \boldsymbol{(x_2 - x_1)\hat{\imath} + (y_2 - y_1)\hat{\jmath}}$$

EXAMPLE : The vector from $\boldsymbol{P_1}(3,4)$ to $\boldsymbol{P_2}(5,1)$ is

$$\overrightarrow{\boldsymbol{P_1P_2}} = (5-3)\hat{\imath} + (1-4)\hat{\jmath}$$
$$= 2\hat{\imath} - 3\hat{\jmath}$$

EXAMPLE : **G**iven the points $P(-2,3)$ and $Q(4,5)$

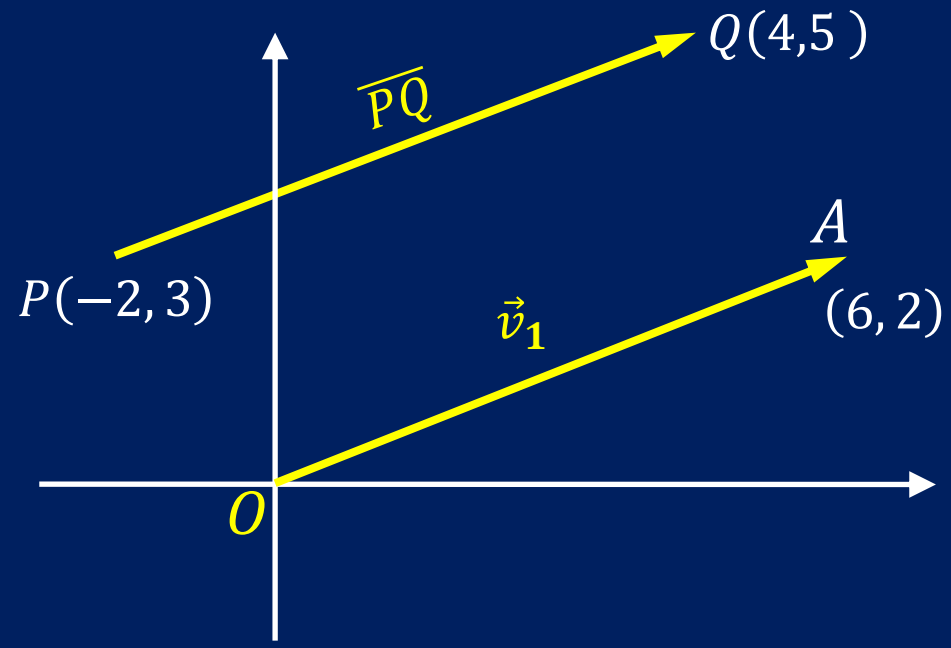
a) Sketch $\overline{PQ}$ and $\overline{QP}$

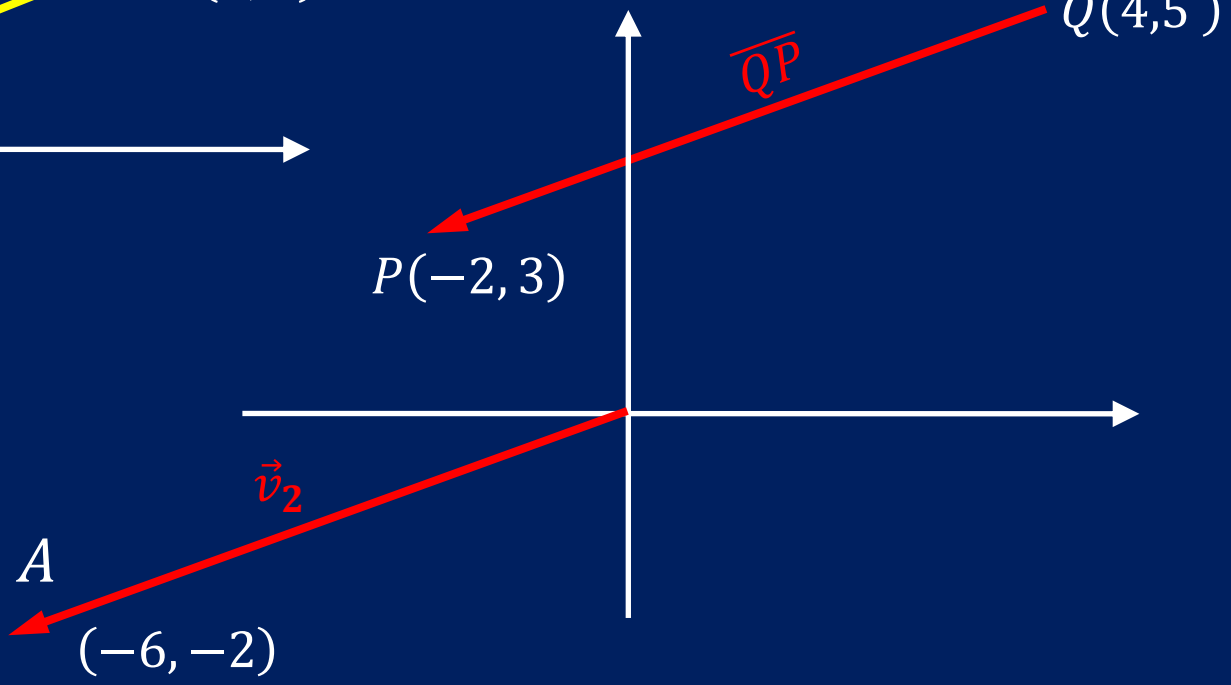
b) Find vectors $\vec{v}_1$ and $\vec{v}_2$ that corresponds to $\overline{PQ}$ and $\overline{QP}$

(Also sketch)

SOLUTION

(a) The Vectors $\overline{PQ}$ and $\overline{QP}$ are sketch in Figure





(b)

$$\overline{PQ} = \vec{v}_1 = (4-(-2))\hat{\imath} + (5-3)\hat{\jmath} = 6\hat{\imath} + 2\hat{\jmath} = (6,2)$$

$$\overline{QP} = \vec{v}_2 = (-2,-4)\hat{\imath} + (3-5)\hat{\jmath} = -6\hat{\imath} - 2\hat{\jmath} = (-6,-2)$$

10) MAGNITUDE OF VECTOR

The magnitude or length of $\vec{v} = a\hat{\imath} + b\hat{\jmath}$ is

$$|\vec{v}| = \sqrt{a^2 + b^2}$$

The bars in $|\vec{v}|$ (read "the magnitude of $\vec{v}$ or the length of $\vec{v}$) are the same bars we use for absolute value.

10) UNIT VECTORS

Any vector whose length is 1 is a unit vector.

The vectors $\hat{\imath}$ and $\hat{\jmath}$ are unit vectors.

$$|\hat{\imath}| = |1\,\hat{\imath} + 0\,\hat{\jmath}| = \sqrt{1^2 + 0^2} = 1$$

$$|\hat{\jmath}| = |0\hat{\imath} + 1\,\hat{\jmath}| = \sqrt{0^2 + 1^2} = 1$$

10) LENGTH vs DIRECTION

If $\vec{v} \neq 0$, then is

1. $\frac{\vec{v}}{|\vec{v}|}$ is a unit vector in the direction of $\vec{v}$.

2. The equation $\vec{v} = |\vec{v}| \left(\frac{\vec{v}}{|\vec{v}|}\right)$ express $\vec{v}$ in terms of its length and direction.

EXAMPLE;

Express $\vec{v} = 3\hat{\imath} - 4\hat{\jmath}$ as a product of its length and direction

Solution;

Length of $\vec{v}$

$$|\vec{v}| = |3\,\hat{\imath} - 4\,\hat{\jmath}| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Direction of $\vec{v}$

$$\frac{\vec{\mathbf{v}}}{|\vec{\mathbf{v}}|} = \frac{3\,\hat{\imath} - 4\,\hat{\jmath}}{5} = \frac{3}{5}\hat{\imath} - \frac{4}{5}\hat{\jmath}$$

so $\vec{v} = 3\hat{\imath} - 4\hat{\jmath} = 5\ (\frac{3}{5}\hat{\imath} - \frac{4}{5}\hat{\jmath})$

Length

Direction

EXERCISE

Q#1 Find a unit vector that has (a) the same direction as $\vec{v}$ and (b) the opposite direction of $\vec{v}$

i) $\vec{v} = -8\hat{i} + 15\hat{j} = (-8, 5)$ $\xrightarrow{\text{Ans}}$ (a) $-\frac{8}{17}\hat{i} + \frac{15}{17}\hat{j}$

(b) $\frac{8}{17}\hat{i} - \frac{15}{17}\hat{j}$

ii) $\vec{v} = 2\hat{i} - 5\hat{j} = (2, 5)$ $\xrightarrow{\text{Ans}}$ (a) $\frac{2}{\sqrt{29}}\hat{i} - \frac{5}{\sqrt{29}}\hat{j}$

(b) $-\frac{2}{\sqrt{29}}\hat{i} + \frac{5}{\sqrt{29}}\hat{j}$

Q#2 Find a vector of magnitude 6 that has the same direction of $\vec{v} = 4\hat{i} - 7\hat{j}$

Ans $\rightarrow$ $\frac{24}{\sqrt{65}}\hat{i} - \frac{42}{\sqrt{65}}\hat{j}$ (using formula $\vec{v} = |\vec{v}|\left(\frac{\vec{v}}{|\vec{v}|}\right)$)

Q#3 Find a vector of magnitude 4 that has the opposite direction of $\vec{v} = 2\hat{i} - 5\hat{j}$

Ans $-\frac{8}{\sqrt{29}}\hat{i} + \frac{20}{\sqrt{29}}\hat{j}$

Q#4 Let $\vec{A} = 2\hat{i} - 7\hat{j}$, $\vec{B} = \hat{i} + 6\hat{j}$ and $\vec{C} = \sqrt{3}\hat{i} - \pi\hat{j}$. Write each of the vectors in the form $a\hat{i} + b\hat{j}$

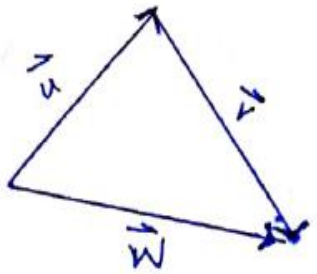
i) $\vec{A} + 2\vec{B}$

(ii) $\vec{A} + \vec{B} - \vec{C}$

(iii) $3\vec{A} - \frac{1}{\pi}\vec{C}$

(iv) $2\vec{A} - 3\vec{B} + 32\hat{j}$

Q#5 Vectors, $\vec{u}$, $\vec{v}$ and $\vec{w}$ are determined by the Sides of triangle ABC as Shown



a) Express $\vec{w}$ in terms of $\vec{u}$ and $\vec{v}$

(b) Express $\vec{v}$ in terms of $\vec{u}$ and $\vec{w}$.

Ans → $\vec{w} = \vec{u} + \vec{v}$

$\vec{v} = \vec{w} - \vec{u}$

Q#6 Express the vectors in the form $a\hat{i} + b\hat{j}$ and sketch them as arrows in the Coordinate plane beginning at the origin.

i) $\overrightarrow{P_1P_2}$ if P_1 is the Point $(5, 7)$ and P_2 is the Point $(2, 9)$.

ii) $\overrightarrow{P_1P_2}$ if P_1 is the Point $(1, 2)$ and P_2 is the point $(-3, 5)$

iii) $\overrightarrow{AB}$ if A is the Point $(-5, 3)$ and B is the Point $(-10, 8)$

iv) $\overrightarrow{AB}$ if A is the point $(-7,-8)$ and B is the Point $(6, 11)$.

v) The sum of the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$, given the four points $A(1,-1)$, $B(2,0)$, $C(-1,3)$ and $D(-2,2)$.

vi) $\overrightarrow{PQ}$ if $P(1,-4)$ and $Q(5,3)$

vii) $\overrightarrow{PQ}$ if $P(-4,6)$ and $Q(-4,-2)$

viii) $\overrightarrow{PQ}$ if $P(-3,-1)$ and $Q(6,-4)$.

Q#7 Given the vector $\overrightarrow{AB} = 3\hat{i} - \hat{j}$ and A is the Point $(2,9)$, find the Point B.

Q#8 Given the vector $\overrightarrow{PQ} = -6\hat{i} - 4\hat{j}$ and Q is the Point (3,3), find the Point P.

Q#9 Express each vector as a product of its length and direction.

(a) $5\hat{i} + 12\hat{j}$ (b) $2\hat{i} - 3\hat{j}$

Q#10 Find the unit vectors that are Parallel to the vector $3\hat{i} - 4\hat{j}$ (two vectors in all).

Q#11 Show that $\vec{A} = 3\hat{i} + 6\hat{j}$ and $\vec{B} = -\hat{i} - 2\hat{j}$ have opposite directions.

Q#12 Show that $\vec{A} = 3\hat{i} + 6\hat{j}$ and $\vec{B} = \frac{1}{3}\hat{i} + \hat{j}$ have the same direction.

Q#13 Let $\vec{A} = 2\hat{i} + \hat{j}$, $\vec{B} = \hat{i} + \hat{j}$ and $\vec{C} = \hat{i} - \hat{j}$.

Find scalars α, β, such that $\vec{A} = \alpha\vec{B} + \beta\vec{C}$

Ans $\alpha = 3/2$

$\beta = \frac{1}{2}$

Q#14 Find all real numbers 'c' such that

(a) $|c\vec{a}| = 3$, (b) $|c\vec{a}| = -3$,

(c) $|c\vec{a}| = 0$.

i) $\vec{a} = 3\hat{i} - 4\hat{j} = (3, -4)$

ii) $\vec{a} = -5\hat{i} + 12\hat{j} = (-5, 12)$

Q#15 Find a vector that has the same direction as $-6\hat{i} + 3\hat{j}$ and

a) twice the magnitude

(b) one-half the magnitude.